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AGENT DESIGN

MAXIT

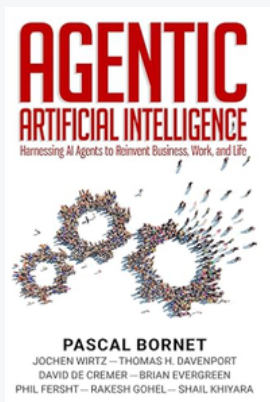
SALES ASSISTANT AGENT FOR CORPORATE & INVESTMENT BANKING



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Acknowledgement



*Agentic Artifical Intelligence
Pascal Bornet et. al.*

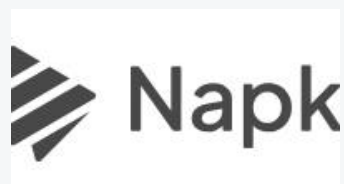


*Philoagents Online Course
Neural Maze & Decoding ML*

My personal assistants



ChatGPT



napkin.ai

buildshed.substack.com



This is a Blueprint to build a “Sales Assistant bot” for Corporate & Investment Banking (CIB)

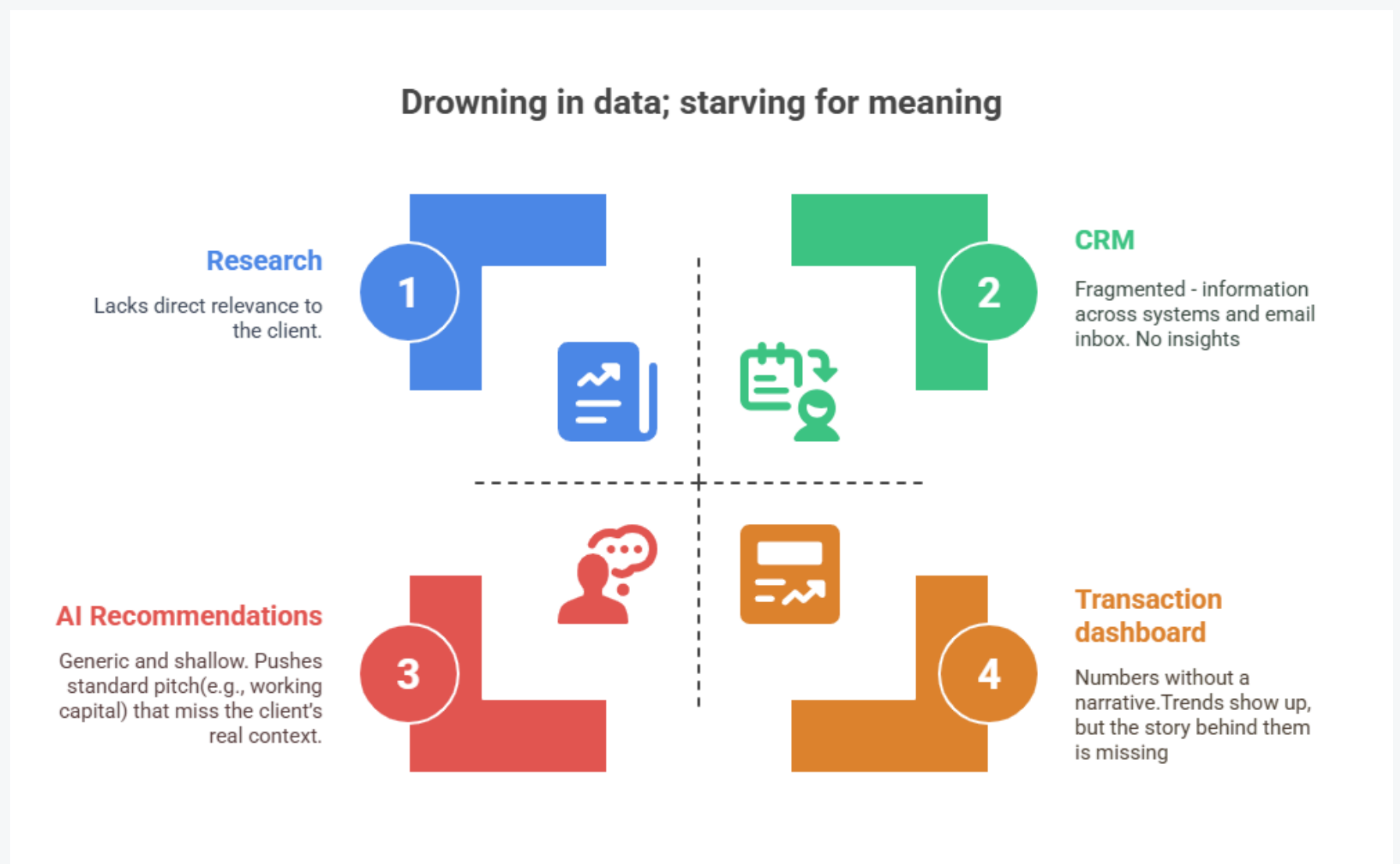
- 1 Problem Statement
.....
- 2 Components
.....
- 3 Illustrations
.....
- 4 Architecture
.....
-



Problem Statement

CIB Relationship Manager:

She has the tools, but not the clarity. Insights are scattered across systems, dashboards, and inboxes—forcing her to piece together context instead of preparing strategy.



Abundant data, fragmented intelligence → heavy reliance on the human to connect the dots and take action



Components

14

components of a “Sales Assistant bot” for CIB RM

Let us explore each component one by one



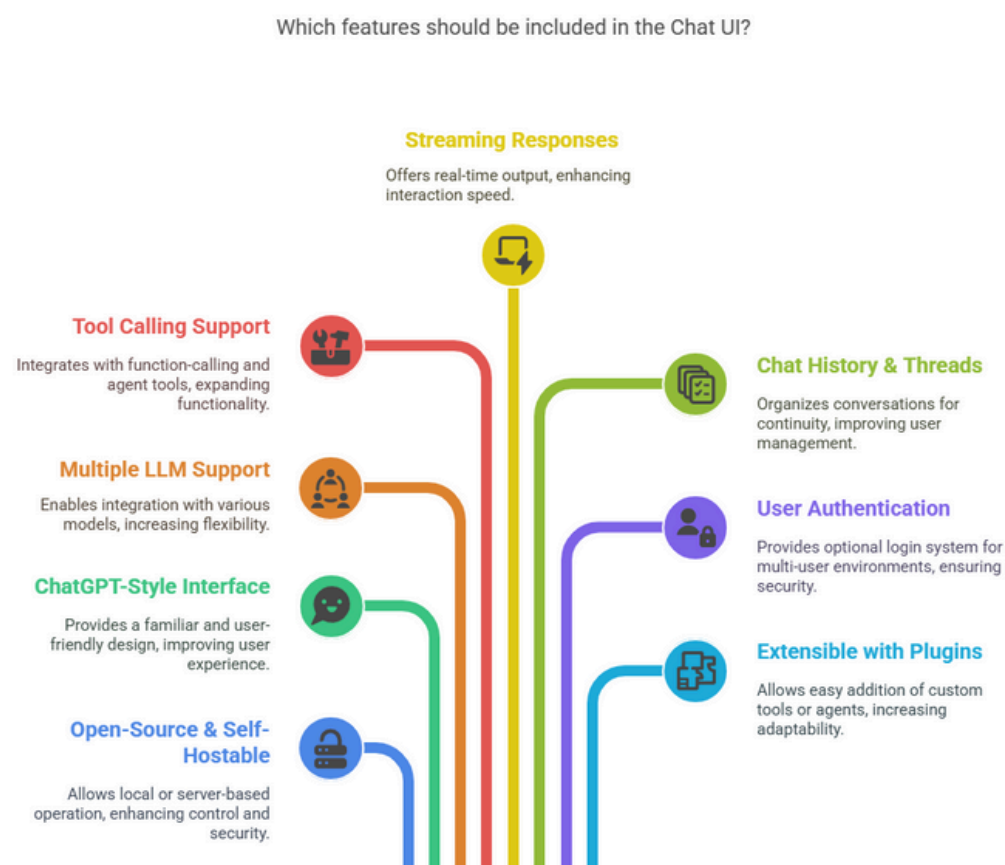
Components

of a “Sales Assistant bot” for CIB RM

#1: Chat UI

WHY? *Do I need a Chat UI*

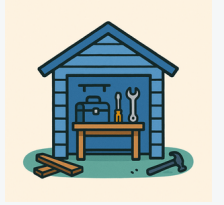
Traditional UIs like CRM are built for data entry and retrieval—not for reasoning, synthesis. Chat interfaces offer a faster, more flexible way to think and act.



✓ Quick Start? Plug in OpenWebUI

⚡ Testing Reasoning Flow? Try LangGraph Agent Chat

🏢 Using Enterprise Chat: Use Google Spaces, Microsoft Teams, or Slack as your ChatUI



Components

#2: Agent Orchestration Framework

WHAT? *is an agent orchestration framework*

An agent orchestration framework manages how tasks flow across tools, memory, and reasoning steps. It defines the logic behind multi-step actions, decisions, and tool use.

WHY? *do I need an agent orchestration framework*

Because solving real problems isn't just about answering a question — it's about managing a conversation, recalling context from memory, and coordinating multi-step logic.

Langgraph and **Llamaindex** are prominent agent orchestration frameworks.



Adopt an agent orchestration framework to accelerate your build.

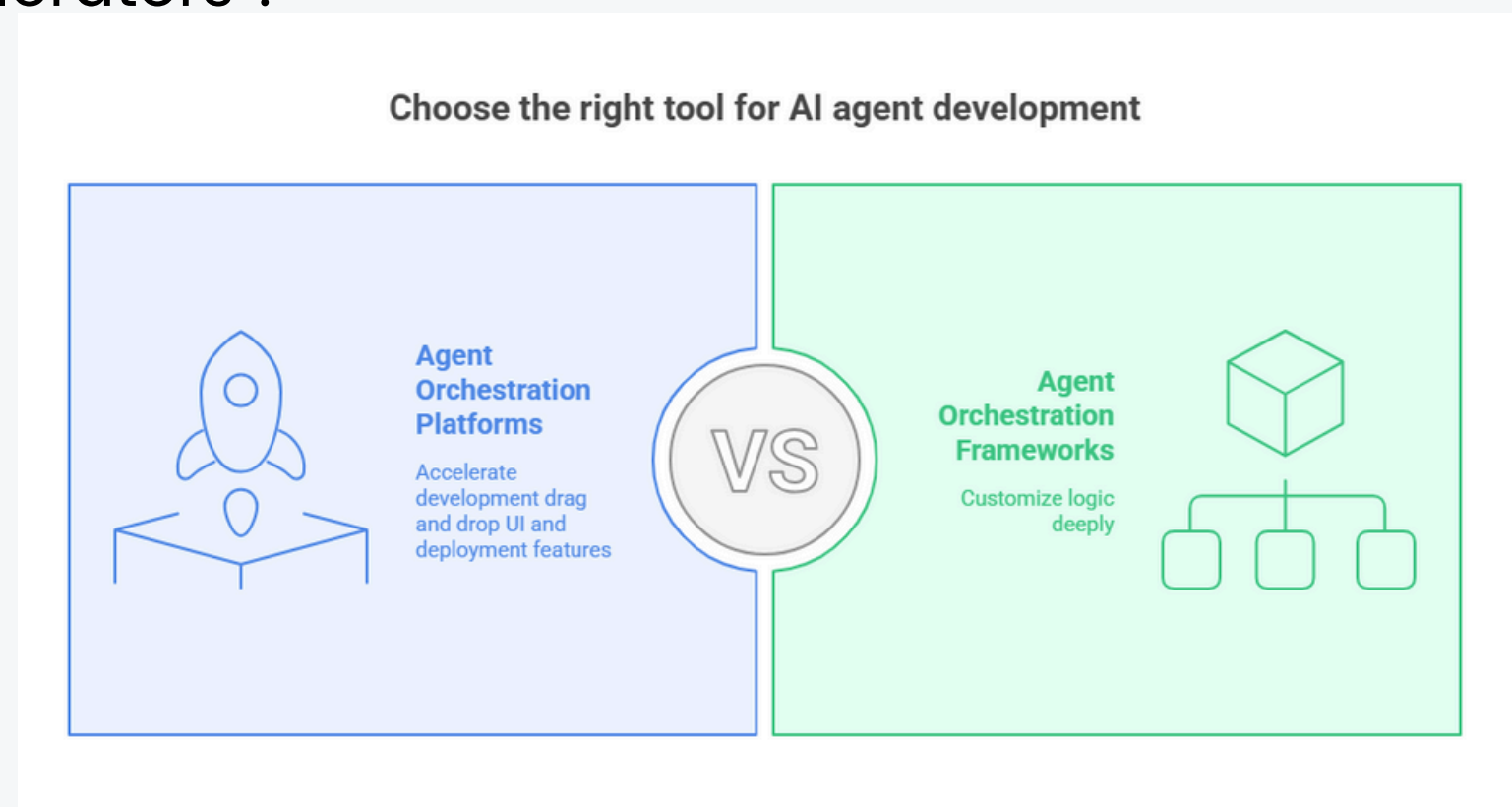


Components

#2: Agent Orchestration Framework

Agent Orchestration Platforms – such as *dify*, *agent vercel*, *n8n* are coming up.

Agent Orchestration platforms have a different intent and design vs. pureplay orchestration frameworks like langgraph. Think of them as “deployment & prototyping accelerators”.



Use orchestration platforms when you need speed and simplicity over full control.



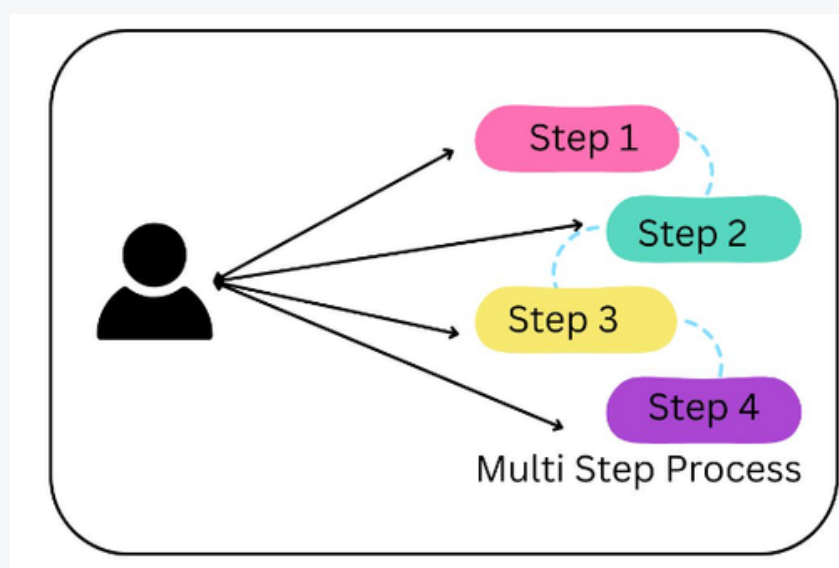
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#3: Agents

WHAT? *is an Agent*

An AI Agent is a goal-driven system that combines reasoning, memory, and tools to take purposeful actions.

WHY? *We have automation tools; so why do we need AI Agents?*



For complex tasks, traditional tools handle parts of the job—leaving the RM to connect the dots, make decisions, and drive the workflow manually.



Components

#3: Agents

How?

is an AI Agent different from automation tools

True AI Agents vs. Automation

Rule based decisions



Adaptive context aware decisions

Fixed tools, fixed flow



Chooses tools, adapts flow.

Comprehensive records, static schema



Fluid evolving structure, contextual retrieval

Rules-Based Automation



True AI Agent

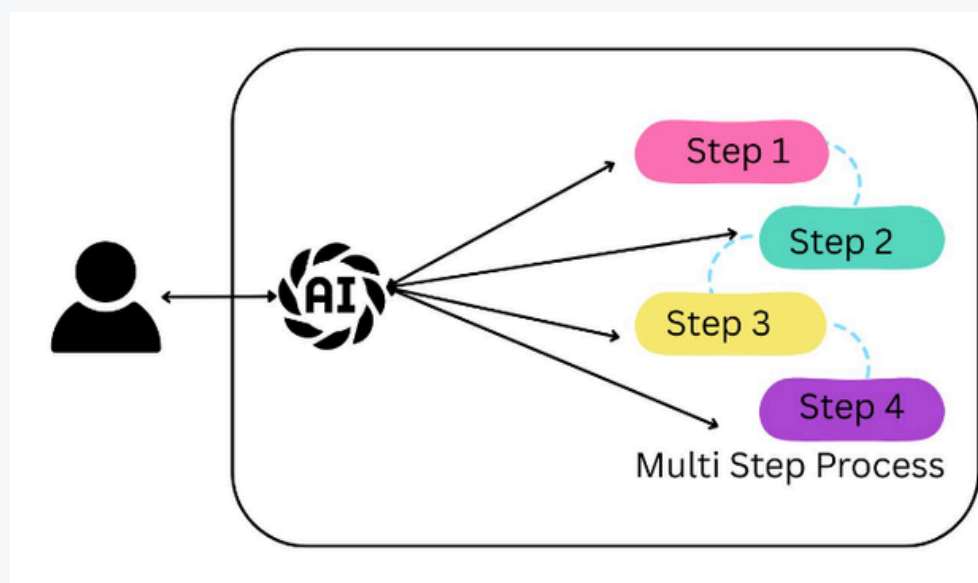


Components

#3: Agents

How?

Can an AI Agent help the RM?



With the Agent, the RM observes and guides. The Agent (LLM) becomes the doer. This elevates the RM from 'grunt work doer' to client partner.

WHAT? *is the Agent design*

A CIB RM's day involves many types of decisions — it's best to think of multiple agents operating in a structure to assist the RM.

For example 'Filings Agent', 'Stock Price Agent'.

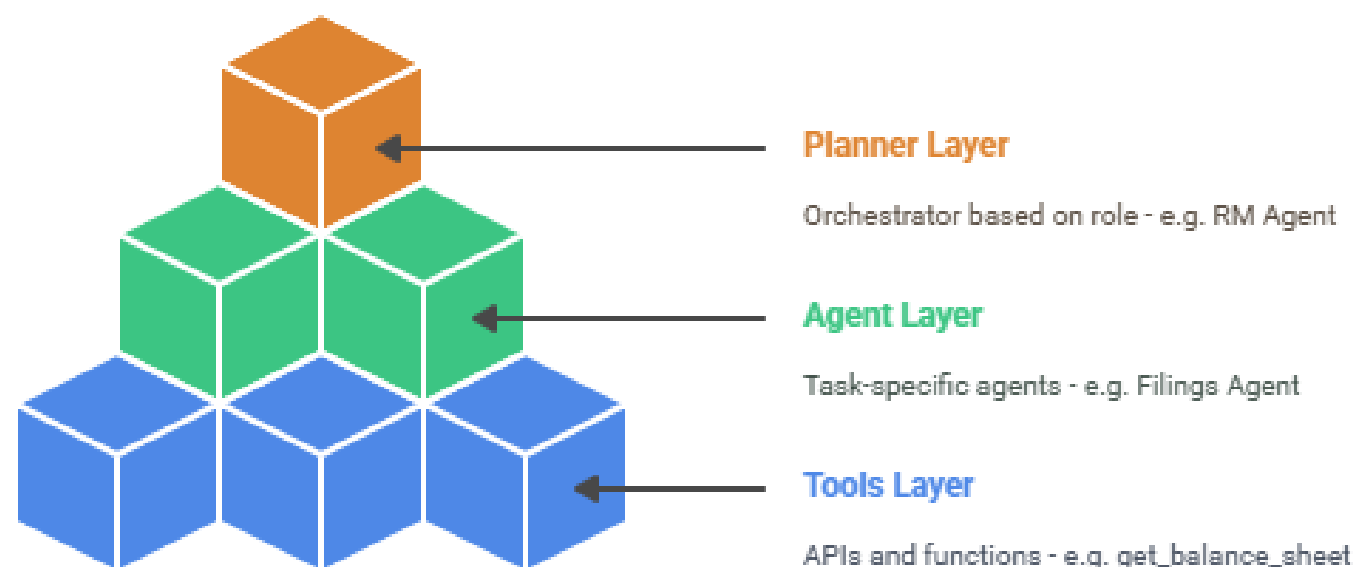


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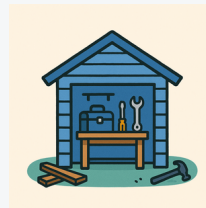
#3: Agents

WHAT? *is the Agent design*

Multi-Layer Agent Architecture



A layered structure of Agents – where Planners guide, Agents specialize, and Tools provide reasoning logic



Components

#4: Planner Agent

WHAT? *is it*

In a multi-agent setup, the Planner acts as a Router — directing each user query to the most relevant agent. E.g. a query about the last annual report goes to 'Filings' agent

WHY? *you need it*

Just like an RM's day — multi-faceted and dynamic — this setup brings the right expertise to each moment..



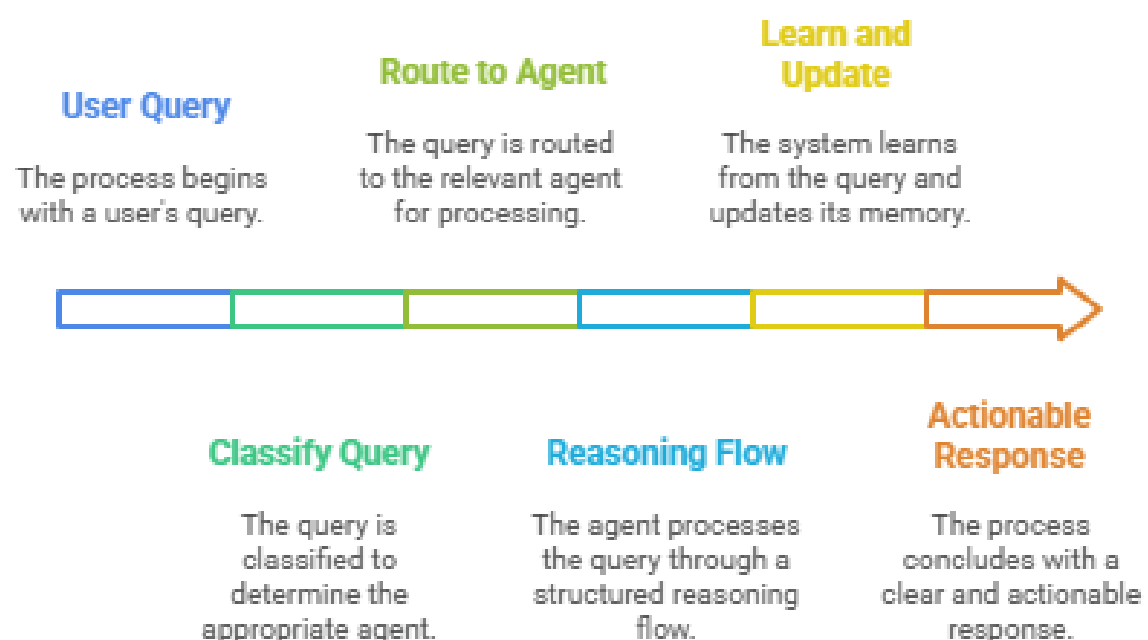
Start with an LLM as the Planner. As you scale, add AI classifiers for more precise routing.



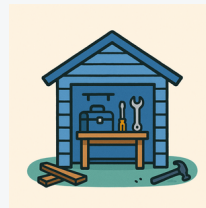
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#5: Reasoning Graph

A structured thinking process —based on the type of question, available data, and desired outcome.



Use a reasoning framework like LangGraph to accelerate your build. It gives structure to agent interactions



Components

#6: Tools

WHAT?

Tools are external functions or APIs an agent can call—like database queries, or calculators. For example *get_balance_sheet (year =2024, company = "Micron")*

WHY?

Do we need Tools?

LLMs have general intelligence, but they don't know your data or systems.

Tools connect the agent to proprietary data and domain-specific logic—making it truly useful in your context.



Components

#6: Tools

Tool categories for Maxit



Break big functions into small, reusable tools—like Lego bricks. Fine-grained tools unlock flexibility and create sharper intelligence.



Components

#7: Actions

WHAT?

Actions are how agents create real impact—by triggering downstream processes, sending messages, updating systems, or surfacing results. Actions move the agent from insight to outcome.

WHY?

Do we need Agents to Act?

Because we have a “role reversal” problem with AI

"This role reversal where humans become "the robots" connecting various systems while AI dreams up possibilities - points to a fundamental misalignment in our approach to artificial intelligence. "

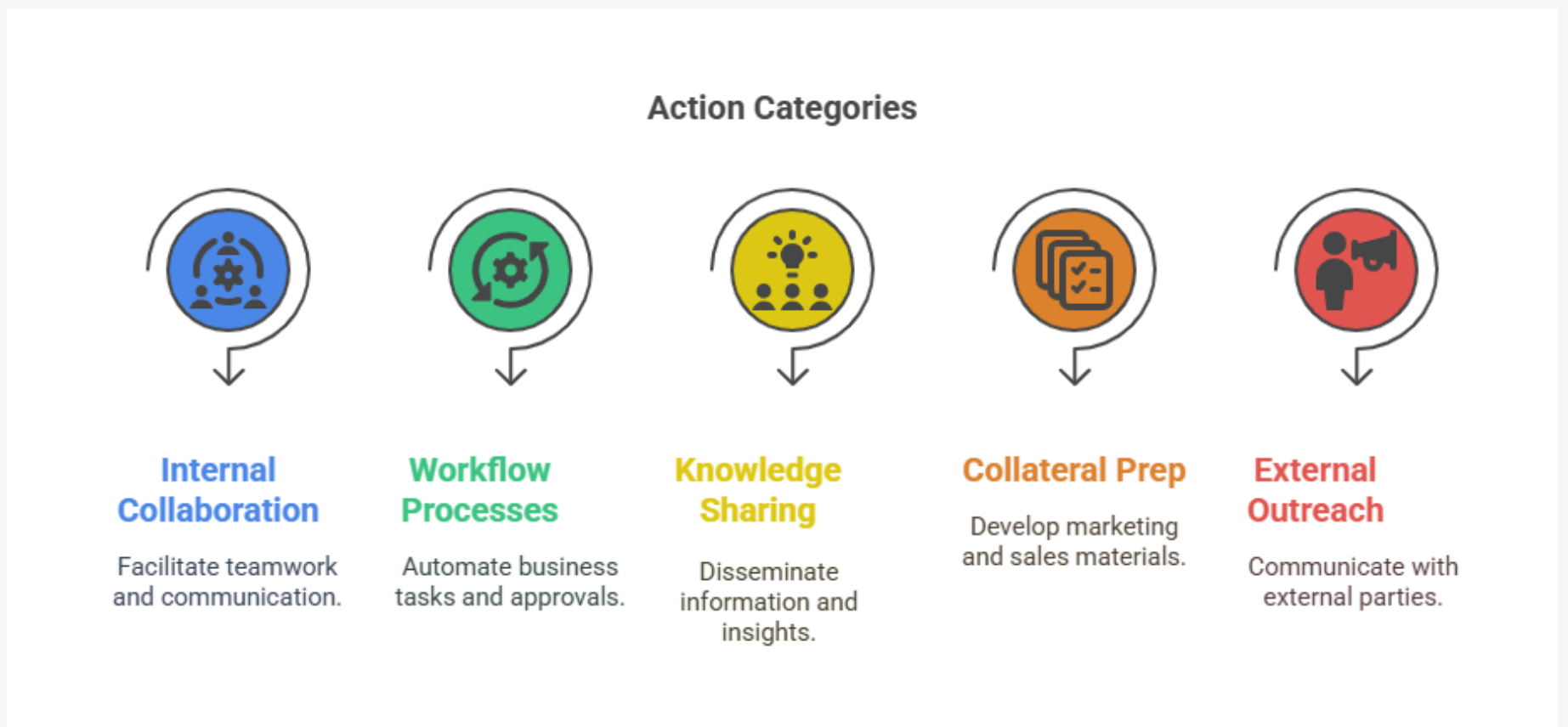
— Agentic Artificial Intelligence, Pascal Bornet et al.



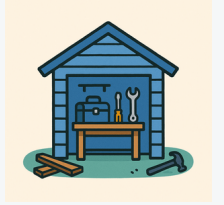
Components

#7: Actions

What actions can Maxit take?



Start from the left (Internal Collaboration).
Build confidence and then move to the right.



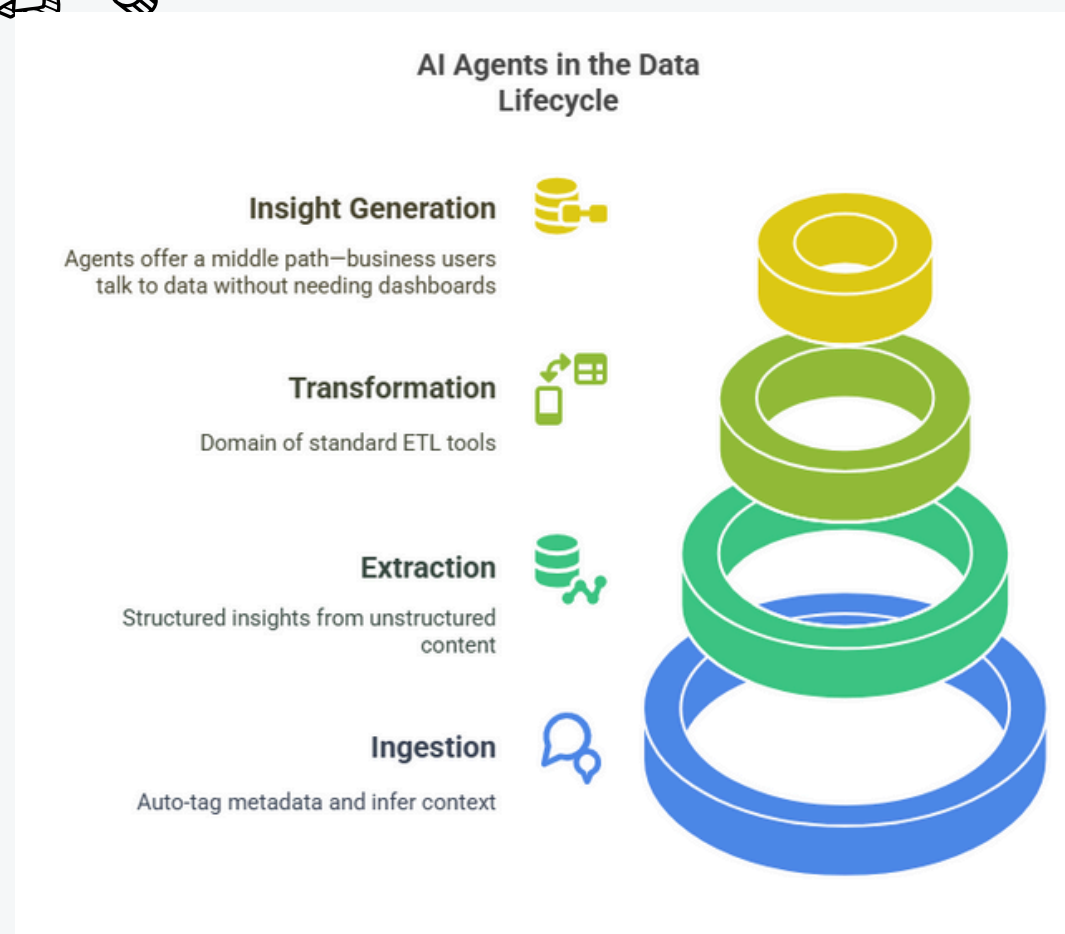
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#8: Data

WHY? *is data critical for AI agents*

Data is central to agent intelligence—it fuels reasoning, decision-making, and action.

WHAT? *Can AI agents do in the data lifecycle*



TOP TIPS



While AI Agents can value add along the full data lifecycle, I recommend to start with “Extraction” first and then “Insight Generation”.



Components

#8: Data

Illustrative data sources for Maxit

RM Assistant Bot Data Sources

Characteristic	Internal	External
 Client Information	Portfolio details, txn history	Filings, investor reports, ownership
 Communication Records	Meeting notes, call reports	News, M&A rumors, public statements
 Research	Macro & sector views	Analyst reports, credit/equity ratings
 Risk Assessment	Credit memos, KYC, internal ratings	Sanctions, credit ratings
 Sales Information	Pitches, deal pipeline, wallet share, coverage themes	Market trends, wallet size

TOP
TIPS



Begin with a blend of internal and external data, and include both slow-moving (e.g. filings) and fast-moving (e.g. market prices) sources.

Don't wait for perfect pipelines — cobble together insights using available tools and data to prototype value fast.



Components

#9: Memory

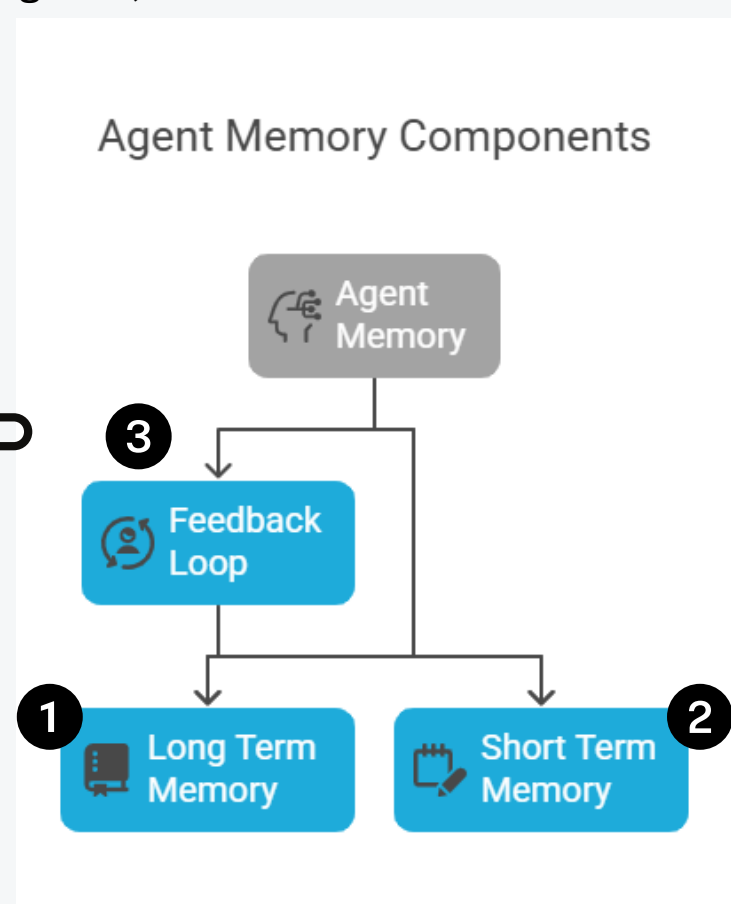
Memory isn't just a feature of intelligence. Memory is the foundation upon which all meaningful intelligence is built

Memory isn't about storing everything, it is about storing what matters.

Memory is more about making meaningful connections than storing perfect records.

— Agentic Artificial Intelligence, Pascal Bornet et al.

Agent Memory has 3 key components





Components

#9: Memory - Long Term



do agents need Long Term memory

Agents lack task persistence by default — they forget what happened prior to this chat.

Long-Term Memory lets the RM assistant stay in sync with ongoing deals, client context, and user preferences, enabling more consistent and proactive support.



to design agent LT memory

6 key questions to answer

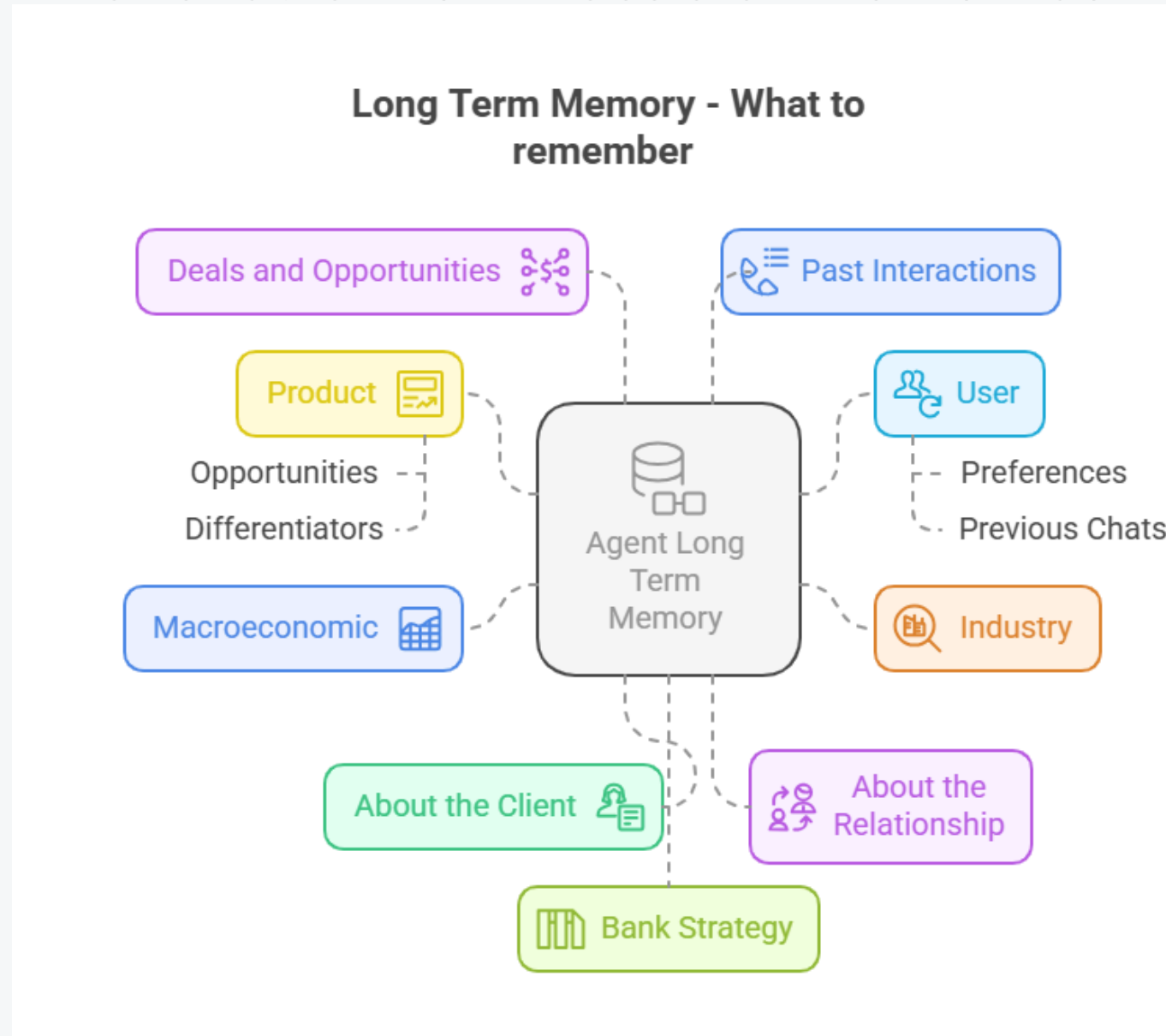




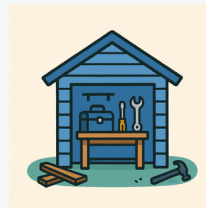
Components

#9: Memory - Long Term

What should the RM assistant remember



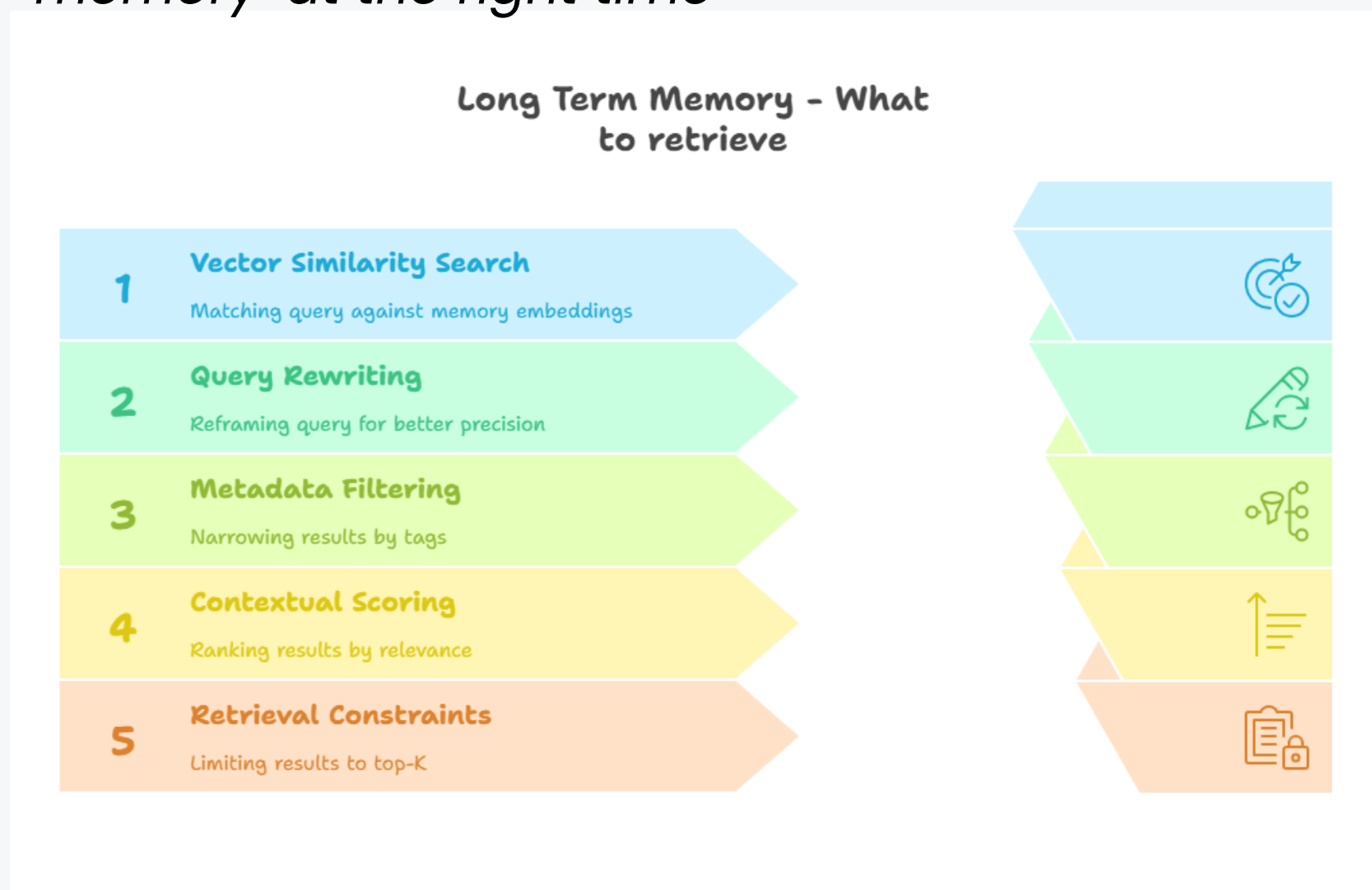
Start with Client, User and Sales (Deals, Opportunities, past interactions).



Components

#9: Memory - Long Term

How does the RM assistant retrieve the right 'memory' at the right time



Start with Metadata filtering and Vector Similarity Search tools



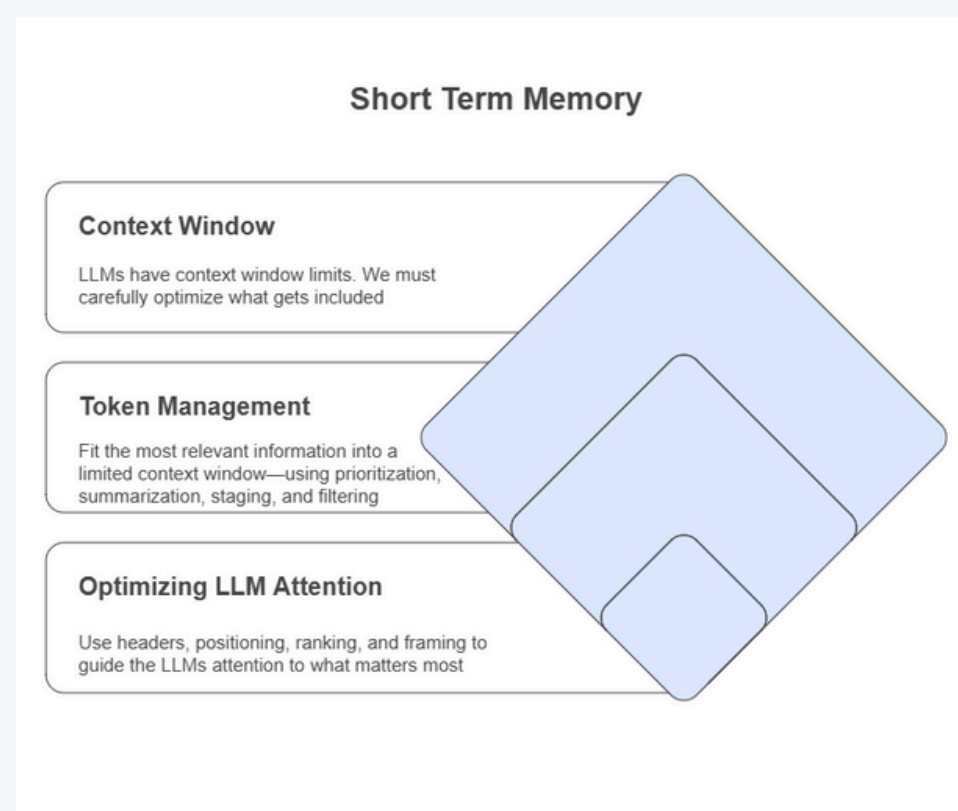
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#9: Memory - Short Term

 Short-Term Memory is about selecting the right slice of history for this moment.

- There's lots of Long-Term Memory (past chats, filings, notes, deals).
- But the LLM's context window is limited — you can't fit everything in.
- So you must filter and extract only what's relevant to the current task.

Short Term Memory Management

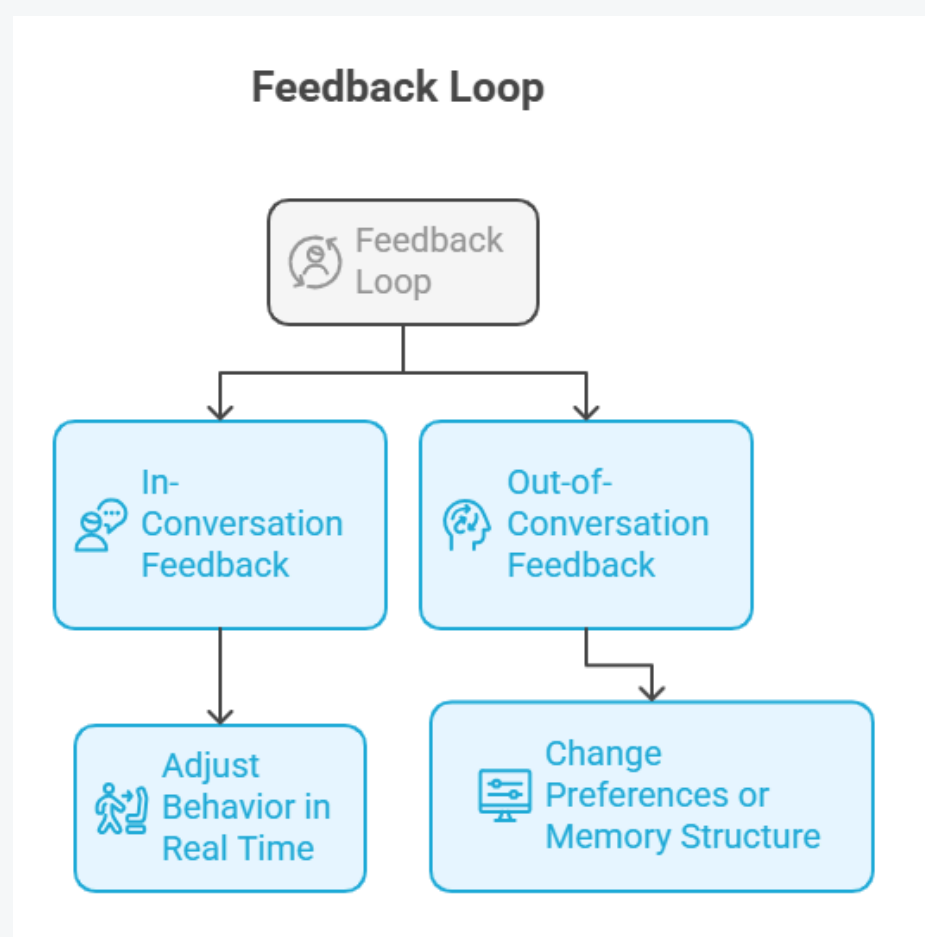


Score chunks for relevance.
Use LLMs to add structure—
like headers and tags



Components

#9: Memory - Feedback Loop

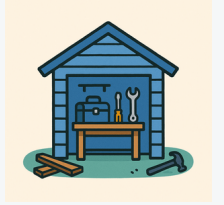


Adjust agent behavior in real-time (in-conversation)
OR
Evolve memory and preferences over time.



In-conversation feedback: Let the LLM handle it to begin with.

Out-of-conversation feedback: Start by editing procedural logic and auto-prompt changes



Components

#10: Human Oversight

WHY? *Do we need human oversight?*

In complex, high-stakes flows like RM assistance, human oversight is essential—for judgment, nuance, and steering the agent when the path isn't clear-cut.

HOW? *to implement human in the loop*

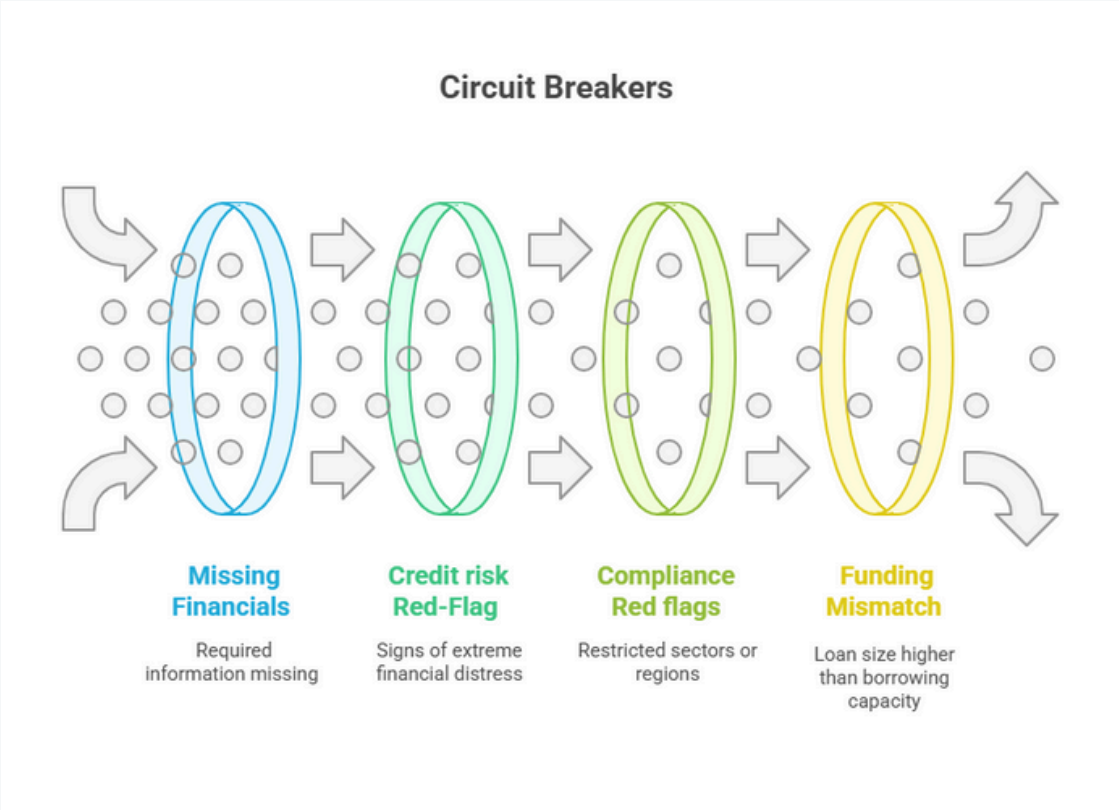
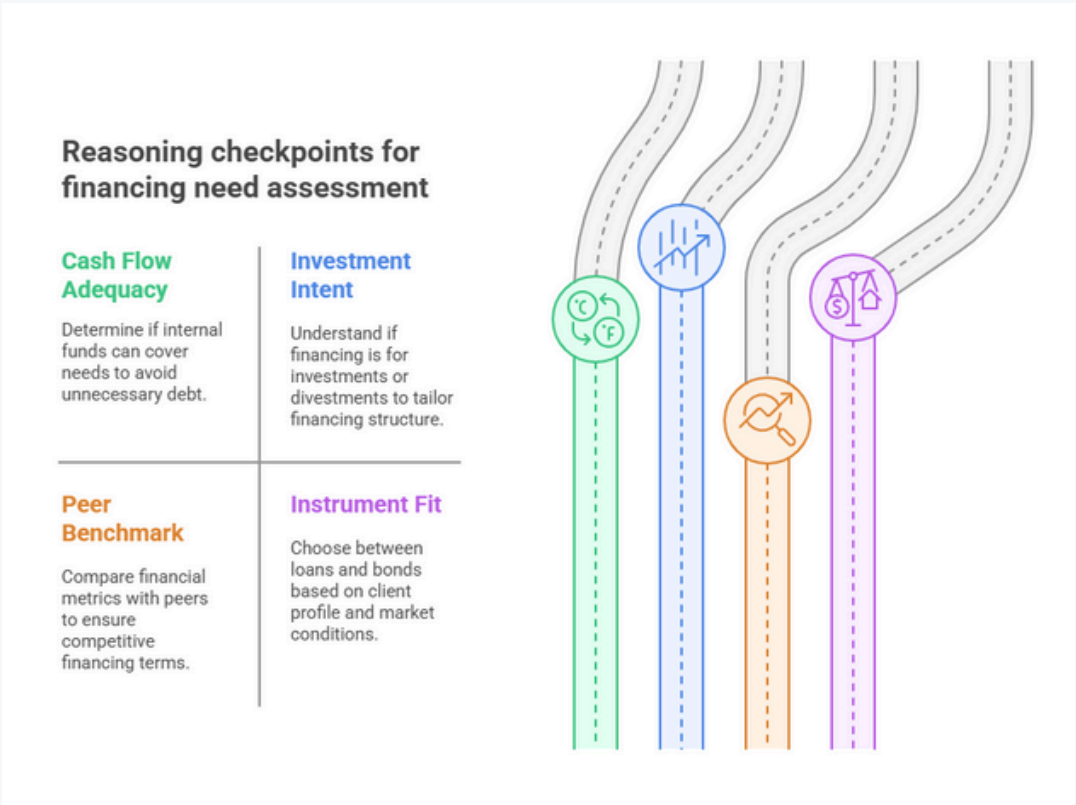




Components

#10: Human Oversight

Reasoning
Checkpoint examples



Circuit Breaker
examples





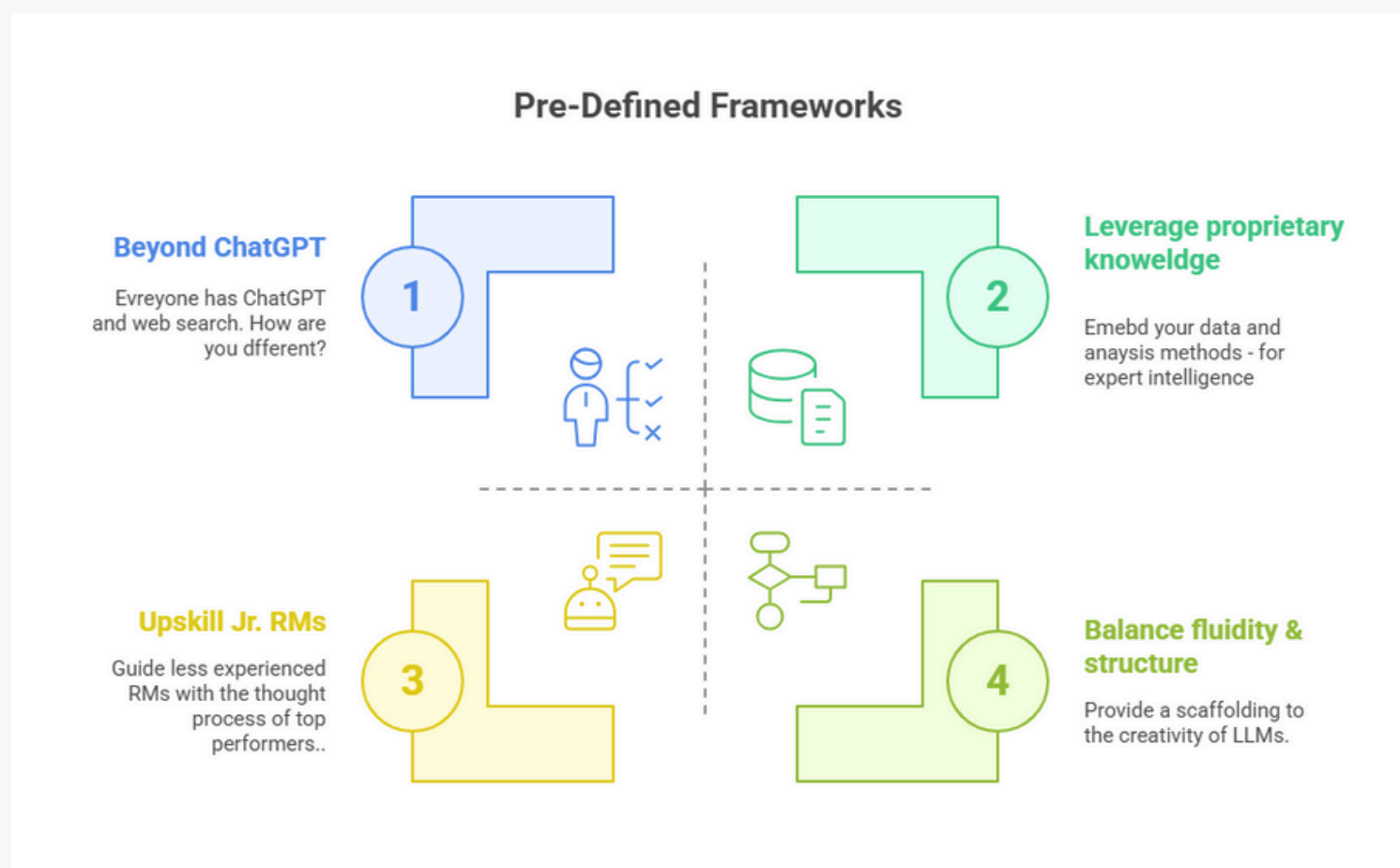
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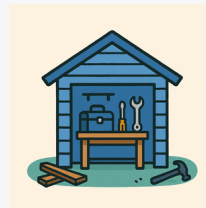
#11: Pre-defined Frameworks

WHAT? *Is a pre-defined framework*

A pre-defined framework is a reusable reasoning pattern tailored to a specific task—like recommending financing.

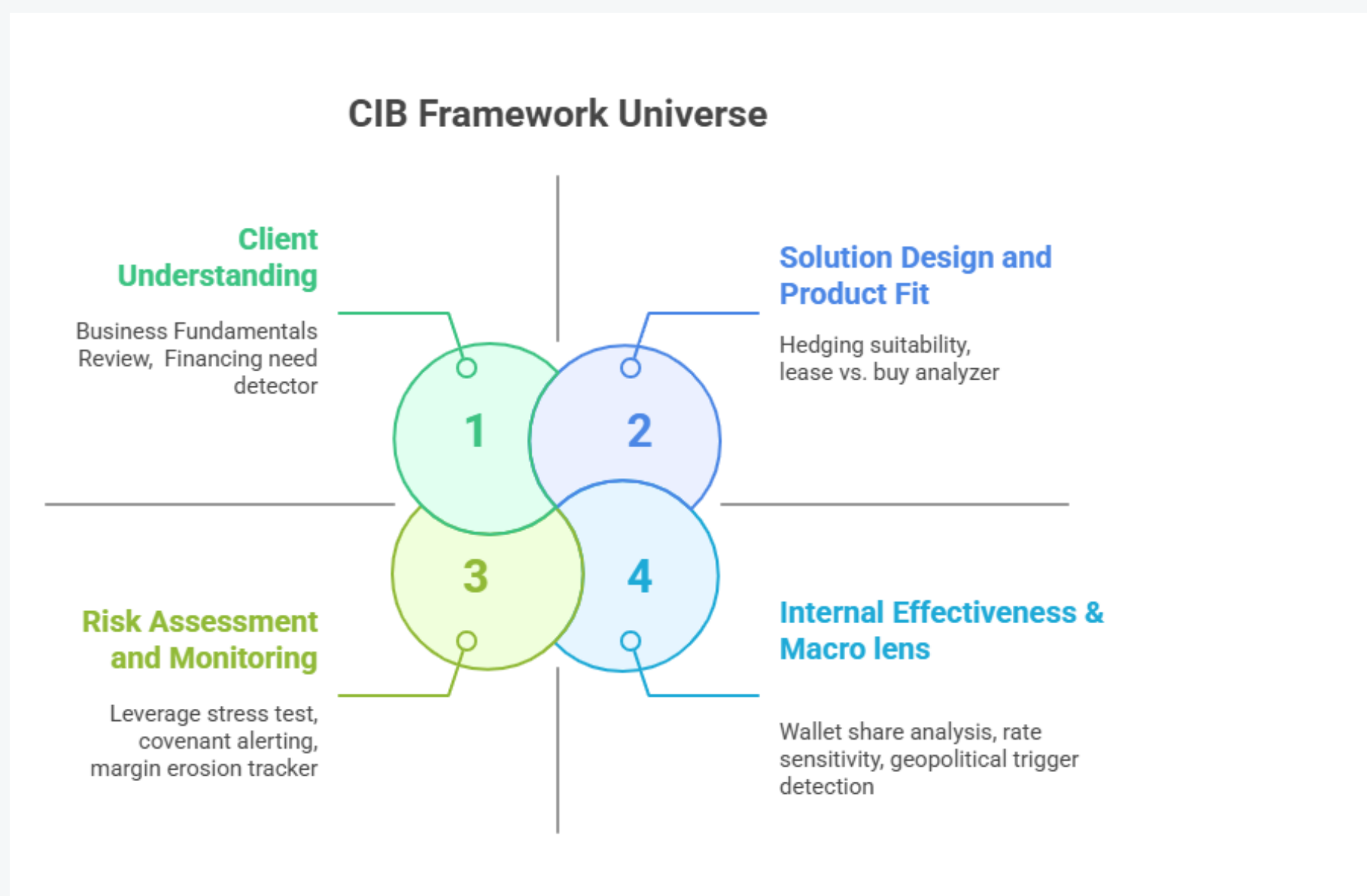
WHY? *do you need pre-defined frameworks? Here are 4 reasons*



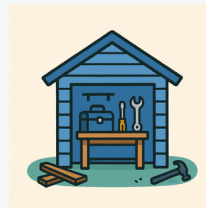


Components

#11: Pre-defined Frameworks



Design frameworks to flex—different teams solve the same task differently (e.g., revenue logic in TMT vs. Industrials). Embed human oversight through reasoning checkpoints and circuit breakers.



Components

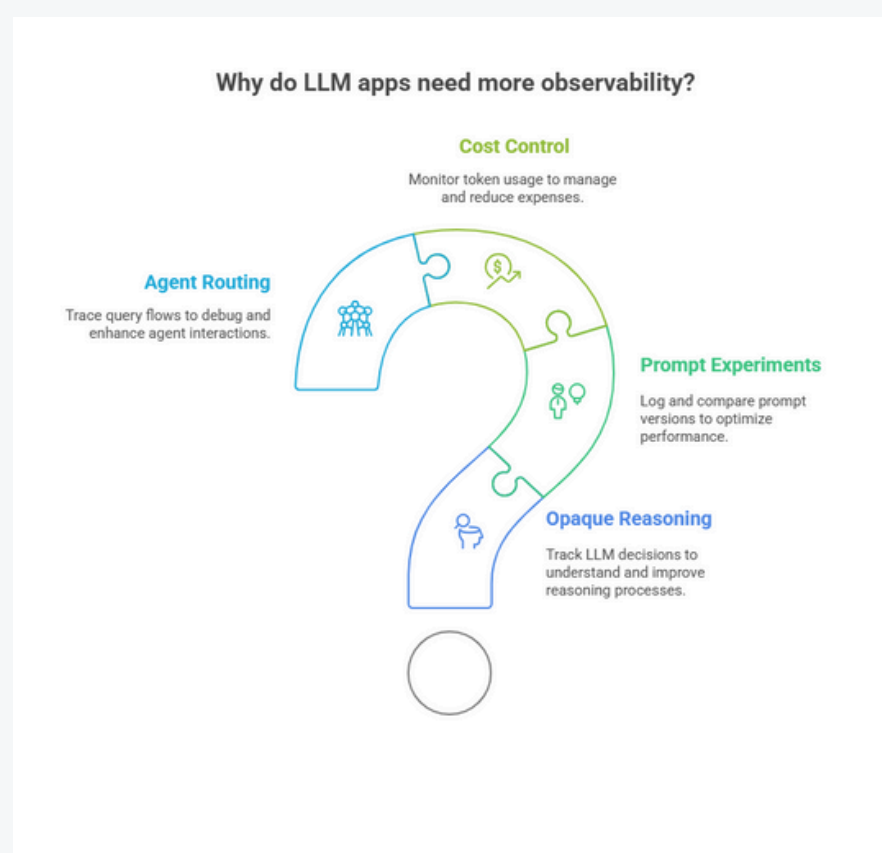
#12: Observability

WHAT?

Observability is the ability to monitor, understand, and debug how your AI system behaves—across inputs, reasoning, outputs, and costs.

WHY?

AI agents need more observability than traditional apps



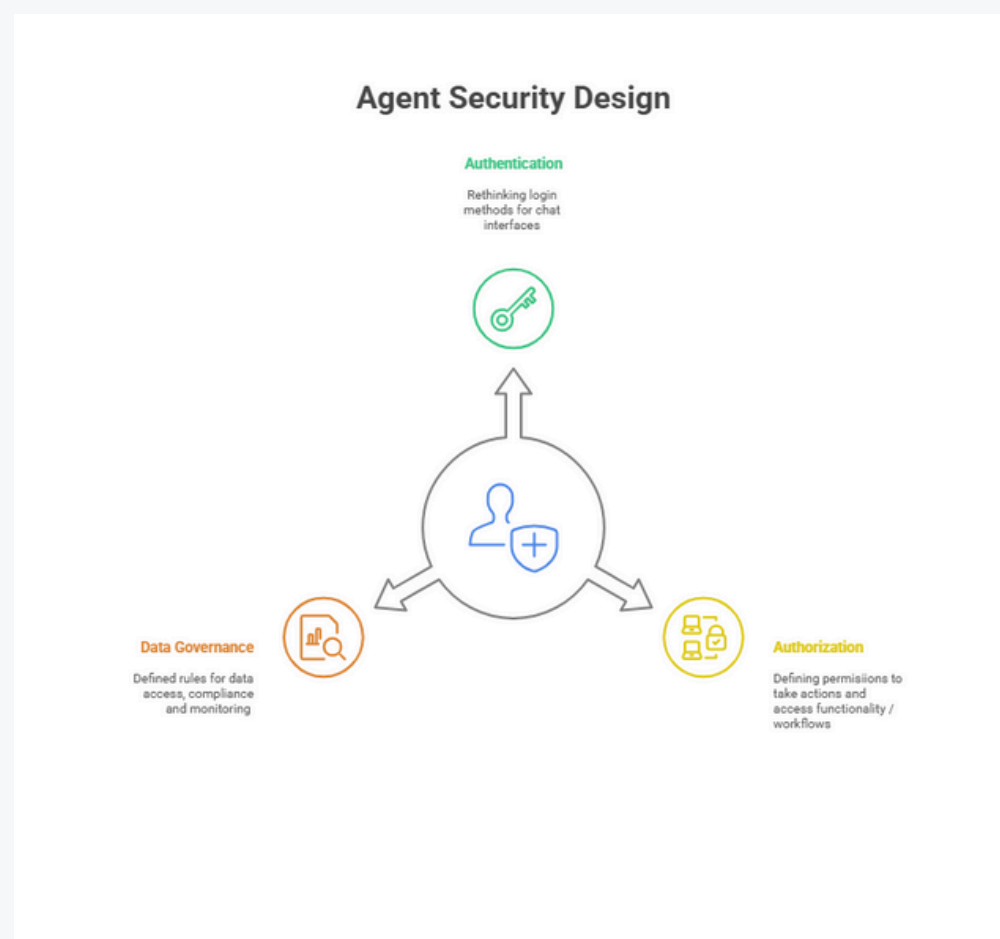
Start with cloud-native logging and tooling.
Use a tool like Opik to get immediate visibility into prompts, flows, and costs.



Components

#13: Security & Data Governance

Conversational Interfaces bring new challenges for Authentication, Authorization and Data Governance



While small-scale pilots are tempting, build in modern auth and data access controls early—it's the foundation that makes scaling safe and sustainable.



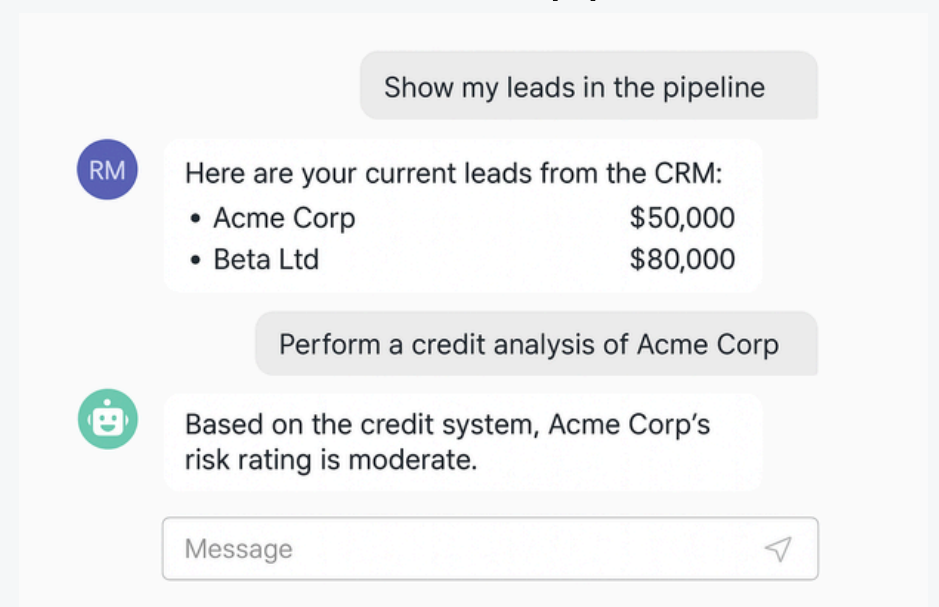
Components

#13: Security & Data Governance

How will Application Security change with AI Agents

1. Conversational interfaces blur traditional app boundaries

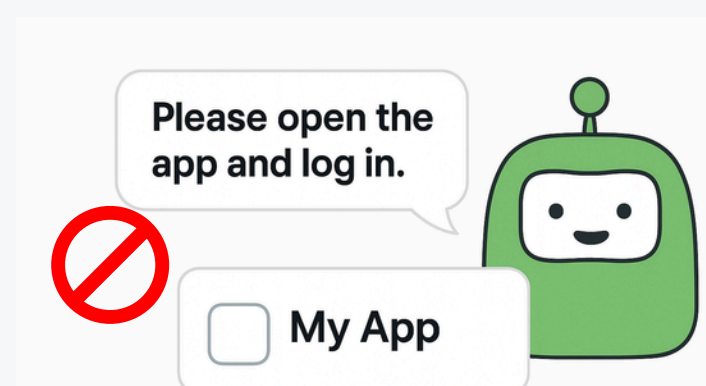
Agents span multiple systems - for example CRM and Credit system.



2. UI-based authentication methods (like login forms) don't work in a chat context



Retrofitting modern token based auth into legacy apps can be complex. Start selectively - CRM and credit systems, for example.





Components

#13: Security & Data Governance

How will Application Security change with AI Agents

3. AI Agent is a reasoning actor, not a static script

	
BEFORE (Static Automation)	NOW (Agents as Reasoning Actors)
• Access is debated, decided, and hard-coded. • Flows are linear Permissions are baked into the script or UI roles. • Rarely re-evaluated once deployed.	• Agent decides dynamically what to do—based on inputs, memory, tools. • You can't predefine every path—it's non-linear, adaptive. • So you must give it leeway—but wrap that leeway in structure.



Security framework – Set rules for who can use what app, trigger which tools, and access which data

Agent as user – Think of agent as a user, with their own permissions.



Components

#14: Evaluation

7 Criteria for Agent Evaluation



Use a tool like Opik to accelerate setup and get immediate visibility into prompts, flows, and costs.

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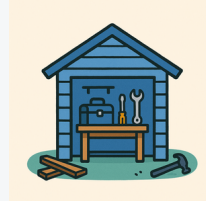


Illustrations

**COMING
SOON**



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Architecture

**COMING
SOON**

